HW 5.1: In this question we will explore one last aspect of internal corrections to the Hydrogen atom (hyperfine splitting, HFS), as well as the response of the atom to an external magnetic field (the Zeeman effects). We work in natural units, so $\hbar = c = 1$ and $\alpha = \frac{e^2}{4\pi} \simeq \frac{1}{137}$ in what follows.

Previously, we considered the total angular momentum and its z projection (j, m_j) as good quantum numbers for the Hydrogen atom and its fine structure perturbations. There, the spin-orbit coupling dependend on $\mathbf{L} \cdot \mathbf{S}$, which commuted with L^2, S^2, and J^2, but not with L_z and S_z.

In this first problem, we will consider the hyperfine structure of the Hydrogen atom. We will tackle the case $\ell \neq 0$ where we will need to account for the magnetic field induced by the orbital motion of the electron or nucleus. (The $\ell = 0$ case is solved in Griffith's book, and he only needs to account for the interaction between spins!) Note that this is harder than our previous S.O. problem: before, we had a dipole (the electron with a spin) interacting with the magnetic field generated by a moving charge (the spinless proton). Now, we have two dipoles (the electron and the nucleus with their spins) interacting with each other, so we have a 1) spin-spin coupling, 2) the proton-spin-electron-orbit coupling, and finally 3) the old electron-spin-proton-orbit coupling we calculated. The latter is unchanged, but the first two are new and we will calculate them.

We will interpret the problem as the interaction of the nuclear spin $\mathbf{I}$ with two internal magnetic fields calculated at the location of the nucleus: 1) the B-field produced by the electron spin (a dipole), and 2) the B-field produced by the orbital motion of the electron:

$$H_{\text{hfs}} = -\mathbf{M}_I \cdot \mathbf{B}_{\text{el}} = -\mathbf{M}_I \cdot \left(\mathbf{B}_{\text{el}}^{\ell} + \mathbf{B}_{\text{el}}^{s}\right), \tag{1}$$

where $\mathbf{M}_I$ is the nuclear magnetic moment and $\mathbf{B}_{\text{el}}$ is the magnetic field produced by the electron at the location of the nucleus (sorry, my boldface μ does not work for some reason, so we will go with $\mathbf{M}_I$ instead). Similarly to what we argued in class, the magnetic field produced by the orbital motion of the electron is given by

$$\mathbf{B}_{\text{el}}^{\ell} = -\mathbf{v} \times \mathbf{E} = \frac{\mathbf{E} \times \mathbf{p}}{m_e} = -\frac{1}{Ze}\frac{dV(r)}{dr}\frac{\mathbf{r} \times \mathbf{p}}{m_e r} = -\frac{\alpha}{e m_e}\frac{\mathbf{L}}{r^3}, \tag{2}$$

where we used the fact that $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, $V(r) = -\frac{Z\alpha}{r}$, and $\mathbf{F} = +Ze\mathbf{E} = -\nabla V(r) = -\frac{dV(r)}{dr}\hat{\mathbf{r}}$ (note this now has a positive charge and the radial coordinate is the same we are used to, namely, $\mathbf{r}$ points from the nucleus to the electron). The magnetic field produced by the electron spin is given by the standard formula for the magnetic field of a dipole:

$$\mathbf{B}_{\text{el}}^{s} = \frac{1}{4\pi}\left[\frac{1}{r^3}\left(3(\mathbf{M}_e \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{M}_e\right) + \frac{8\pi}{3}\mathbf{M}_e \delta^3(\mathbf{r})\right] = -\frac{g_e \alpha}{2 e m_e}\left[\frac{1}{r^3}\left(3(\mathbf{S} \cdot \hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{S}\right) + \frac{8\pi}{3}\mathbf{S}\delta^3(\mathbf{r})\right], \tag{3}$$

where $\mathbf{M}_e = -\mu_B g_e \mathbf{S} = -g_e \frac{e}{2m_e}\mathbf{S}$ is the magnetic moment of the electron, μ_B is the Bohr magneton, and g_e is the electron g-factor. The first term is a standard result in E&M (the field of a dipole looks like a quadrupole from far away) and the second accounts for what happens right at the location of the origin (if we think of spin as an infinitesimal current loop, then the magnetic field at the center of the loop blows up, hence the delta function term). This delta function term is called the Fermi contact term. The factor of g_e is the electron g-factor, which we said was $\simeq 2$.

Putting everything together, the HFS perturbation is given by[1]

$$H_{\rm hfs} = -\mathbf{M}_I \cdot \mathbf{B}_{\rm el} = \frac{\alpha g_I}{2m_N m_e}\mathbf{I}\cdot\mathbf{A} \text{ where } \boxed{\mathbf{A} \equiv \frac{\mathbf{L}}{r^3} + \frac{g_e}{2}\left(\frac{1}{r^3}\left(3(\hat{\mathbf{r}})(\mathbf{S}\cdot\hat{\mathbf{r}}) - \mathbf{S}\right) + \frac{8\pi}{3}\mathbf{S}\delta^3(\mathbf{r})\right)}. \tag{4}$$

We used the nuclear magnetic moment $\mathbf{M}_I = +g_I\frac{e}{2m_N}\mathbf{I}$, where g_I is the nuclear g-factor and m_N is the nucleon mass (there is no Z factor here because g_I is the g-factor of a composite state and not the result of directly summing the moment of Z protons or N neutrons). In what follows, you may assume that our nucleus has spin $I = 1/2$ (this is the case for the proton, but not for all nuclei).

Now, we define the total angular momentum of the system as

$$\mathbf{F} = \mathbf{I} + \mathbf{J} = \mathbf{I} + \mathbf{L} + \mathbf{S},$$

where we expect the new good quantum numbers to be (n, l, j, I, f, m_f).

a) Which of the following operators does the HFS perturbation commute with: J^2, J_z, S^2, S_z, I^2, I_z, F^2, F_z? (Hint: do not try to prove it, just use what I told above about what is a good quantum number and what is not.)

b) Starting from the irreps of the three angular momentum operators $\mathbf{S}$, $\mathbf{L}$, and $\mathbf{I}$, construct the irreps of $\mathbf{F}$. In other words, decompose $\mathbf{2}\otimes\mathbf{2}\otimes(\mathbf{2l+1})$ into direct sums of irreps of $\mathbf{F}$. What is the range of values that f can take for a given j and I? What about m_f? What are the $n = 1$ states (previously, the degenerate ground states)?
(Hint: recall that in the spin-orbit problem, we had to decompose $\mathbf{2}\otimes\mathbf{2}$ into irreps of $\mathbf{J}$, and we found that j could take values $|l-s|$ to $l+s$.)

c) Calculate the total HFS energy correction for the $\ell = 0$ states.
(Hint: for $\ell = 0$, use the fact that the wavefunctions are spherically symmetric and express $\mathbf{I}\cdot\mathbf{S}$ in terms of operators that commute with $\mathbf{F}$.)

d) Calculate the HFS energy correction for the $\ell \neq 0$ states.
(Hint: for $\ell \neq 0$, the wavefunctions are not spherically symmetric and we have no choice but to calculate the expectation value of some nasty operators. Thankfully, the Wigner-Eckart theorem comes to our aid with a very useful relation:

$$\mathbf{A} \longrightarrow \frac{(\mathbf{A}\cdot\mathbf{J})}{j(j+1)}\mathbf{J} \tag{5}$$

where we have used the fact that $\mathbf{J}$ are the only vectors that we can construct from $\mathbf{A}$ that have the right transformation properties under rotations, and we have fixed the proportionality constants by taking the dot product of both sides with $\mathbf{J}$. You might also benefit from

$$(\mathbf{S}\cdot\hat{\mathbf{r}})^2 = S_i S_j \hat{r}_i\hat{r}_j = \frac{1}{4}(\sigma_i\sigma_j)\hat{r}_i\hat{r}_j = \frac{1}{4}\left(\delta_{ij} + i\epsilon_{ijk}\sigma_k\right)\hat{r}_i\hat{r}_j = \frac{\hat{\mathbf{r}}\cdot\hat{\mathbf{r}}}{4} = \frac{1}{4}, \tag{6}$$

where we used the fact that $\epsilon_{ijk}\hat{r}_i\hat{r}_j = 0$ since $\hat{r}_i\hat{r}_j$ is symmetric in i and j, while ϵ_{ijk} is antisymmetric. You will also need the integral $\langle\frac{1}{r^3}\rangle$ for the hydrogenic wavefunctions:

$$\left\langle\frac{1}{r^3}\right\rangle = \int d^3r\,|\psi_{n\ell m}(\mathbf{r})|^2\frac{1}{r^3} = \frac{1}{n^3 a_0^3}\frac{1}{\ell(\ell+1/2)(\ell+1)} = \frac{(Z\alpha)^3 m_e^3}{n^3}\frac{1}{\ell(\ell+1/2)(\ell+1)}, \tag{7}$$

)

[1] Note the absence of the Thomas precession factor of 1/2 in this calculation since we have not changed frames; before it arose because the electron rest frame was not inertial.

e) Express the HFS energy for the ground state of Hydrogen (the 1S state) in eV and then in Hz. What photon energy would correspond to this transition in centimeters? In what frequency band does this transition lie? Which of the following drawings on the voyager plate is alluding to this transition?

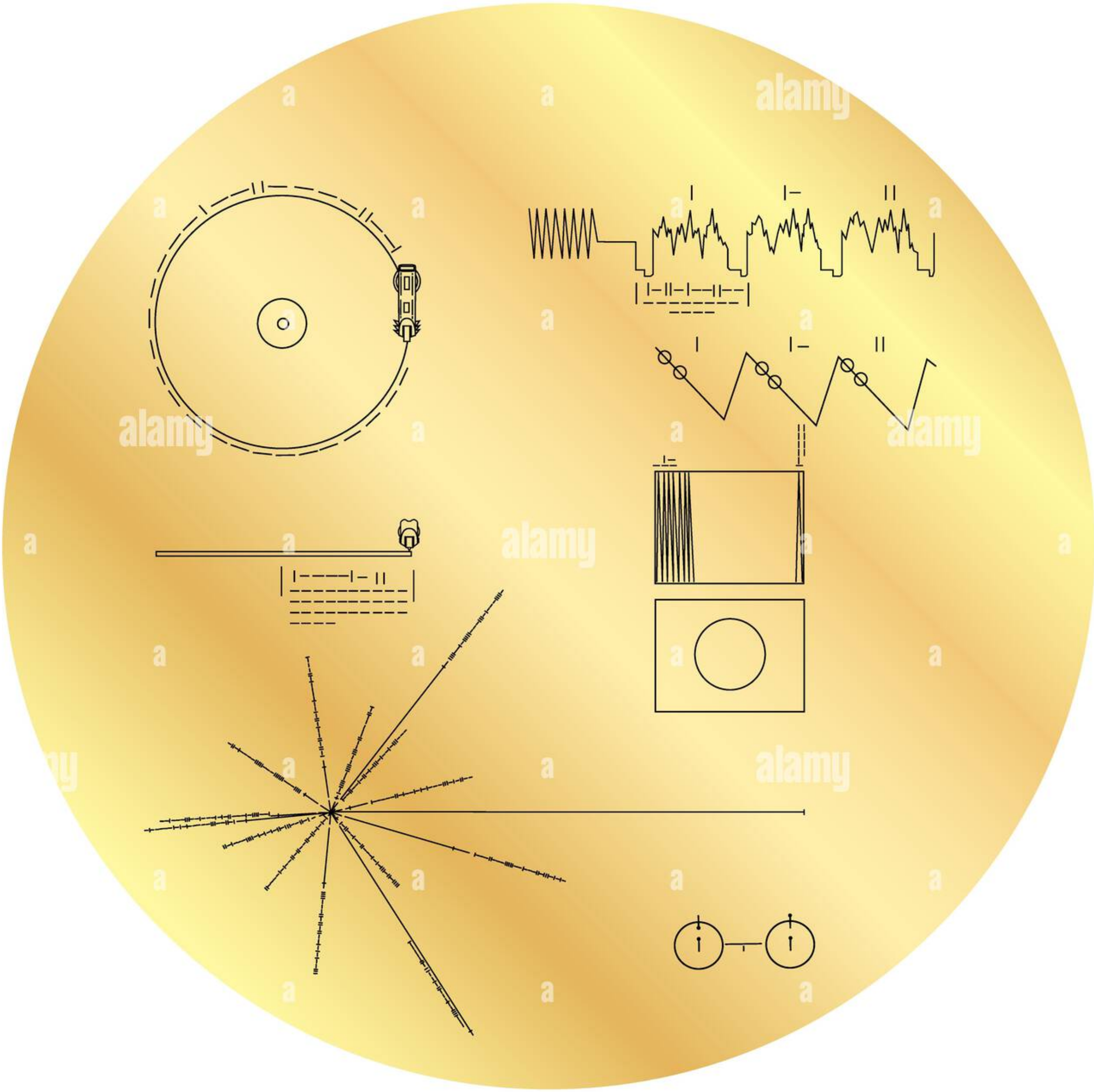

Figure 1: The cover of the Voyager Golden Record (reproduction of the original record sent to space in 1977).

HW 5.2: Consider now an external magnetic field $\mathbf{B}_{\text{ext}} = B\hat{\mathbf{z}}$. The Zeeman effect is the splitting of energy levels due to the interaction of the magnetic field with the magnetic moments of the electron and nucleus. For now, let us ignore the nuclear spin and focus on the interaction of the magnetic field with the electron.

Including the electron-spin-orbit coupling, the total Hamiltonian of the system is given by

$$\hat{H}_{\text{Zeeman}} = \hat{H}_0 - \mathbf{M}_e \cdot \left(\frac{\mathbf{B}_{\text{int}}}{2} + \mathbf{B}_{\text{ext}}\right) - \mathbf{M}_\ell \cdot \mathbf{B}_{\text{ext}} = \hat{H}_0 + \frac{g_e \alpha}{4 m_e^2 r^3} \mathbf{L} \cdot \mathbf{S} + \frac{e}{2m_e}(L_z + g_e S_z)B. \quad (8)$$

where we used $\mathbf{M}_e = -g_e \frac{e}{2m_e}\mathbf{S}$ with $g_e = 2$ and $\mathbf{M}_\ell = \frac{e}{2m_e}\mathbf{L}$ (note the 1/2 in the internal field to account for the Thomas precession factor).

a) Which of the following operators does the Zeeman perturbation commute with: J^2, J_z, S^2, S_z, L^2, L_z?

b) Calculate the correction to the energy levels due to the Zeeman effect in the weak field limit (where the Zeeman perturbation is smaller than the fine structure spin-orbital terms).
(Hint 1: first find the good quantum numbers in this case).
(Hint 2: again, use $S_z \longrightarrow \frac{(\mathbf{S} \cdot \mathbf{J})}{j(j+1)} J_Z$ and the fact that $j = l \pm \frac{1}{2}$, unless $\ell = 0$, when $j = \frac{1}{2}$.
)

c) Calculate the correction to the energy levels due to the Zeeman effect in the strong field limit, neglecting the spin-orbit term.
(Hint: Recall that angular momentum conservation is a consequence of spherical symmetry, so if we break it, we should not expect to have good quantum numbers associated with J^2, for example. Instead, L_z and S_z will be good quantum numbers since the system is still symmetric under rotations around the z-axis.)

d) What happens in the intermediate field limit? What would be your plan of attack to solve the problem in this case? (Hint: do not try to do it.)

Solutions

HW 5.1 Step by step below, including plenty of discussion. This is a long problem. You can find more details in here: `https://bohr.physics.berkeley.edu/classes/221/1112/notes/hyperfine.pdf`.

The original calculations goes back to Fermi, and you can find it here:

Fermi, E. Über die magnetischen Momente der Atomkerne. Z. Physik 60, 320-333 (1930). https://doi.org/10.1007/BF01339933

You can compare our Hamiltonian to his Eq (26).

a) From the statement of the question, f and m_f are good quantum numbers, so we expect the HFS perturbation to commute with F^2 and F_z. Similarly for I^2, S^2, and J^2 since we are told that i, s, and j are good quantum numbers as well. On the other hand, I_z, S_z, and J_z are not good quantum numbers, so we the HFS perturbation do not commute with them (this is intuitive, since the HFS perturbation contains terms that look like I_xS_x and I_yS_y, which do not commute with I_z and S_z).

Proof of commutation relations This was beyond what I was hoping we needed to do, but here it goes (thanks Esme and Halit for bringing up the issues with calculating commutators with J^2). The HFS perturbation contains the scalar product of $\mathbf{I}$ with $\mathbf{L}$ as well as with $\mathbf{S}$. If we can find which of the operators these scalar products commute with, then we can find which operators the HFS perturbation commutes with. A few relations you should know are:

$$[I_i, S_j] = [I_i, L_j] = [S_i, L_j] = 0 \quad \forall i, j, \tag{9}$$
$$[S_i, S^2] = [J_i, J^2] = [L_i, L^2] = [I_i, I^2] = [F_i, F^2] = 0, \tag{10}$$

The first is a consequence of the fact that each angular momentum operator acts on a different space (namely, the nuclear spin, electron spin, and orbital spaces). The second is easy to show by explicit calculation. You may also memorize it as a manifestation of the fact that "we can only simultaneously diagonalize the total angular momentum and one of the individual angular momenta (but not all of them at the same time)". We are free to choose which one, namely, which i we want. In group theory, one may say that the square of an angular momentum operator is a Casimir operator of the rotation group, and therefore commutes with all generators of the group.

Equipped with this knowledge, we can easily check that the following is true:

$$[\mathbf{I}\cdot\mathbf{L},\, S^2] = 0, \tag{11}$$
$$[\mathbf{I}\cdot\mathbf{L},\, I^2] = 0, \tag{12}$$
$$[\mathbf{I}\cdot\mathbf{S},\, S^2] = 0, \tag{13}$$
$$[\mathbf{I}\cdot\mathbf{S},\, I^2] = 0. \tag{14}$$

Good, but we are not done as we also need to check the commutation with the J and F operators. One can show that

$$[\mathbf{I}\cdot\mathbf{L},\, J^2] = [\mathbf{I}\cdot\mathbf{L},\, L^2 + S^2 + 2\mathbf{L}\cdot\mathbf{S}] = 2[\mathbf{I}\cdot\mathbf{L},\, \mathbf{L}\cdot\mathbf{S}] \tag{15}$$
$$= 2I_iS_j[L_i, L_j] = 2i\epsilon_{ijk}I_iS_jL_k = 2i(\mathbf{I}\times\mathbf{S})\cdot\mathbf{L}, \tag{16}$$

as well as

$$[\mathbf{I}\cdot\mathbf{S},\, J^2] = [\mathbf{I}\cdot\mathbf{S},\, L^2 + S^2 + 2\mathbf{L}\cdot\mathbf{S}] = 2[\mathbf{I}\cdot\mathbf{S},\, \mathbf{L}\cdot\mathbf{S}] \tag{17}$$
$$= 2I_iL_j[S_i, S_j] = 2i\epsilon_{ijk}I_iL_jS_k = 2i(\mathbf{I}\times\mathbf{L})\cdot\mathbf{S}, \tag{18}$$

so direct computation here has failed. One may hope these two pieces cancel in the full perturbation, but this is not immediately obvious since the coefficients in front of $\mathbf{I}\cdot\mathbf{L}$ and $\mathbf{I}\cdot\mathbf{S}$ are different in the way we wrote it. The commutators with F^2 suffer from the same issue.

There is a more general way of proving this, however, which is to invoke the fact that $\mathbf{A}$ is a vector operator that knows only about L and S, and therefore, by the Wigner-Eckart theorem, it must be proportional to $\mathbf{J}$ when acting on the irreps of $\mathbf{J}$. In other words, it is J who generates the rotations of $\mathbf{A}$. This is why we can write $\mathbf{A}\longrightarrow\frac{(\mathbf{A}\cdot\mathbf{J})}{j(j+1)}\mathbf{J}$, as we will do later.

Let us carry out this substitution already:

$$H_{\text{hfs}}=\frac{\alpha g_I}{2m_N m_e}\mathbf{I}\cdot\mathbf{A}\longrightarrow\frac{\alpha g_I}{2m_N m_e}\frac{(\mathbf{A}\cdot\mathbf{J})}{j(j+1)}\mathbf{I}\cdot\mathbf{J}. \tag{19}$$

In this case, one can show (see part (d) for more details) that

$$\boxed{\mathbf{A}\cdot\mathbf{J}=\frac{1}{r^3}\left(\mathbf{L}^2+\left(1-\frac{g_e}{2}\right)\mathbf{L}\cdot\mathbf{S}\right).} \tag{20}$$

For $g_e=2$, this is just $\mathbf{L}^2/r^3$, which is a scalar operator under rotations that commutes with everything. We then just need to prove that $\mathbf{I}\cdot\mathbf{J}$ also commutes with J^2 and F^2. This is easy to show by direct computation:

$$[\mathbf{I}\cdot\mathbf{J},\,J^2]=[\mathbf{I}\cdot(\mathbf{L}+\mathbf{S}),\,\mathbf{L}^2+\mathbf{S}^2+2\mathbf{L}\cdot\mathbf{S}]=2[\mathbf{I}\cdot\mathbf{L},\,\mathbf{L}\cdot\mathbf{S}]+2[\mathbf{I}\cdot\mathbf{S},\,\mathbf{L}\cdot\mathbf{S}]=0, \tag{21}$$

by virtue of Eqs. (15) and (17). In much a similar way, we can show that

$$[\mathbf{I}\cdot\mathbf{J},\,F^2]=0 \tag{22}$$

and we are done! For a review of the Wigner-Eckart theorem, see Sakurai's book, section 3.11. Here, we have used the Projection Theorem.

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b) The decomposition of the irreps of $\mathbf{S}$, $\mathbf{L}$, and $\mathbf{I}$ into direct sums of irreps of $\mathbf{F}$ can be obtained as a two-step process: first, we decompose $\mathbf{S}$ and $\mathbf{L}$ into irreps of $\mathbf{J}$, and then we decompose the resulting irreps of $\mathbf{J}$ with $\mathbf{I}$ into irreps of $\mathbf{F}$.[2]

The first step is the same as in the spin-orbit problem (stare at this until it makes sense; try out a few examples with different ℓ):

$$2\otimes(2\ell+1)=(2\ell+2)\oplus(2\ell)\longrightarrow j=\begin{cases}\ell\pm\frac{1}{2} & \text{if } \ell\neq 0\\ \frac{1}{2} & \text{if } \ell=0\end{cases} \tag{23}$$

where it is understood that we are dealing with the dimensions of the irreps and not their labels. For example, ℓ is the label of the irrep of $\mathbf{L}$, but the dimension of the irrep is $2\ell+1$, in the same way that $\frac{1}{2}$ is the label of the irrep of $\mathbf{S}$, but the dimension of the irrep is 2. The equation above can be completely equivalently written in terms of the labels as

$$\frac{1}{2}\otimes\ell=(\ell+\frac{1}{2})\oplus(\ell-\frac{1}{2}). \tag{24}$$

Since $s=1/2$ is the only possible value for the electron spin, adding it to any angular momentum can only give you a single half-integer step up or down in units of angular momentum (i.e., $\pm\hbar/2$).

[2]In what follows, I will drop the boldface notation for the irreps as my ℓ boldface is not working and $\mathbf{l}$ is too close to $\mathbf{1}$.

Since the direct product is associative, we can write

$$
\begin{aligned}
2 \otimes [(2\ell+2) \oplus (2\ell)] &= [2 \otimes (2\ell+2)] \oplus [2 \otimes (2\ell)] && (25)\\
&= [(2\ell+3) \oplus (2\ell+1)] \oplus [(2\ell+1) \oplus (2\ell-1)]\,. && (26)
\end{aligned}
$$

In terms of the labels, we have

$$
\begin{aligned}
\frac{1}{2} \otimes \left[(\ell+\frac{1}{2}) \oplus (\ell-\frac{1}{2})\right] &= \left[\frac{1}{2} \otimes (\ell+\frac{1}{2})\right] \oplus \left[\frac{1}{2} \otimes (\ell-\frac{1}{2})\right] && (27)\\
&= [(\ell+1) \oplus \ell] \oplus [\ell \oplus (\ell-1)]\,. && (28)
\end{aligned}
$$

You can deduce these results by just noting that adding a spin-1/2 to an angular momentum ℓ can only give you $\ell \pm 1/2$, which can either increase the dimension by 1 or decrease it by 1. Applying this logic to each of the two irreps of $\mathbf{J}$ we found in the first step gives the desired results.

Note that the possible values of f are given by $|j-i|$ to $j+i$, which for $i = 1/2$ means that f can take values

$$
f = j \pm \frac{1}{2} = \begin{cases} \ell-1,\ \ell,\ \ell+1 & \text{if } \ell \neq 0 \\ 0 \text{ or } 1 & \text{if } \ell = 0 \end{cases} \tag{29}
$$

where we used the fact that j can take values $\ell \pm 1/2$ (or $1/2$ if $\ell = 0$). Finally, m_f can take values from $-f$ to f in integer steps, so the number of states for a given f is $2f+1$. For the $n = 1$ states, we have $\ell = 0$, so $j = 1/2$ and f can take values 0 and 1. Phew!

What does $f = 0$ mean? Clearly, to get such a scenario we must have $j = s = 1/2$ and $i = 1/2$. In this case, we have two spin-1/2 particles, the electron and the nucleus, which are in a spin-singlet state. Their spins are anti-aligned such that the net spin is zero. On the other hand, when $f = 1$ and $\ell = 0$, the electron and nucleus are in a spin-1 configuration (this is called the triplet state since we have three possible m_f values), and their spins are aligned. We may also have anti-aligned spins with $\ell \neq 0$ that give us $f = 1$, but $f = 0$ is only possible if the electron and nucleus are in a singlet state.

c) Let us write the entire HFS perturbation again:

$$
H_{\text{hfs}} = \frac{\alpha g_I}{2m_N m_e} \mathbf{I} \cdot \left(\frac{\mathbf{L}}{r^3} + \frac{g_e}{2} \left(\frac{1}{r^3} \left(3(\hat{\mathbf{r}})(\mathbf{S} \cdot \hat{\mathbf{r}}) - \mathbf{S}\right) + \frac{8\pi}{3} \mathbf{S} \delta^3(\mathbf{r}) \right) \right). \tag{30}
$$

For $\ell = 0$, the first term is clearly zero since $\mathbf{L}$ acting on any state with $\ell = 0$ gives zero, $L^2 |\ell\rangle = \ell(\ell+1)|\ell\rangle = 0$ and $L_z |\ell\rangle = 0$. The second term is a tensor operator contracted on two vector operators, $\mathbf{I}$ and $\mathbf{S}$. It is easier to work with

$$
T_{ij} = 3\hat{r}_i \hat{r}_j - \delta_{ij}, \tag{31}
$$

such that the second term can be written as $T_{ij} I_i S_j$. Integrating over the angular variables is then relatively straightforward in the $\ell = 0$ case:

$$
\int d\Omega\, T_{ij} = 3 \int d\Omega\, \hat{r}_i \hat{r}_j - 4\pi\delta_{ij} = 4\pi\delta_{ij} - 4\pi\delta_{ij} = 0, \tag{32}
$$

where we used the fact that if $r_i \neq r_j$, then $\hat{r}_i \hat{r}_j$ is an odd function and integrates to zero. For $\hat{r}_i^2$, we have that the integration on the area of the sphere is $\int d\Omega\, \hat{r}^2 = 4\pi$, but by $r^2 = r_1^2 + r_2^2 + r_3^2 = 1$ and by rotational symmetry, each of the three terms must contribute equally, so $\int d\Omega\, \hat{r}_i^2 = 4\pi/3$.

Griffith's asks you to prove it is the case with the relation

$$\int d\Omega\,(\mathbf{a}\cdot\hat{\mathbf{r}})(\mathbf{b}\cdot\hat{\mathbf{r}}) = \frac{4\pi}{3}\mathbf{a}\cdot\mathbf{b}, \tag{33}$$

from which it follows that $\langle 3(\mathbf{I}\cdot\hat{\mathbf{r}})(\mathbf{S}\cdot\hat{\mathbf{r}}) - \mathbf{I}\cdot\mathbf{S}\rangle = 0$. This is equivalent to what we did above.

There is a nice way to understand this result, which is to realize that we are calculating the expectation value of a tensor operator, $T_{ij} = 3\hat{r}_i\hat{r}_j - \delta_{ij}$, in a spherically symmetric state. Therefore, the expectation value must also be a spherically symmetric tensor, but the only spherically symmetric tensor is the identity δ_{ij}, and since T_{ij} is traceless, its expectation value must be zero. A similar argument would apply to the first term, $\mathbf{L}/r^3$, where $\mathbf{L}$ is also a vector operator. In this case, the expectation value must be a spherically symmetric vector, but the only spherically symmetric vector is the zero vector, so the expectation value must also be zero. In this case it obviously does not make sense to have non-zero $\mathbf{L}$ for $\ell = 0$, but the argument still holds. This argument fails for the last delta-function term since it is a scalar operator, so it survives the operation of integrating over the angular variables.

Therefore, for $\ell = 0$, the HFS perturbation reduces to

$$H_{\text{hfs}} = \frac{\alpha g_I g_e}{2m_N m_e}\frac{4\pi}{3}(\mathbf{I}\cdot\mathbf{S})\delta^3(\mathbf{r}), \tag{34}$$

Calculating the expectation value on the ground state:

$$\langle H_{\text{hfs}}\rangle = \frac{2\alpha g_e g_I \pi}{3m_N m_e}\langle\mathbf{I}\cdot\mathbf{S}\rangle|\psi(0)|^2. \tag{35}$$

The ground state of Hydrogen wavefunction at the origin is given by $|\psi(0)|^2 = \frac{1}{\pi a_0^3}$, where $a_0 = 1/(\alpha m_e)$ is the Bohr radius, so we have

$$\langle H_{\text{hfs}}\rangle = \frac{2\alpha g_e g_I}{3m_N m_e a_0^3}\langle\mathbf{I}\cdot\mathbf{S}\rangle = \frac{2\alpha^4 m_e g_e g_I}{3}\frac{m_e}{m_N}\langle\mathbf{I}\cdot\mathbf{S}\rangle. \tag{36}$$

Now, all that is left to do is calculate the expectation value of $\mathbf{I}\cdot\mathbf{S}$ in the ground state. For $\ell = 0$, we have $\mathbf{F} = \mathbf{I} + \mathbf{S}$, therefore,

$$\langle\mathbf{I}\cdot\mathbf{S}\rangle = \frac{1}{2}\langle F^2 - I^2 - S^2\rangle = \frac{1}{2}\left(f(f+1) - i(i+1) - s(s+1)\right), \tag{37}$$

noting we are working with the eigenstates $|n,\ell,j,i,f,m_f\rangle$, so we can replace the operators by their eigenvalues. We can already use the fact that $s = 1/2$ and $i = 1/2$ to simplify the expression above. In particular, when $\ell = 0$, we have $f = 0$ or $f = 1$, so that

$$\langle I.S\rangle_{f=1} - \langle I.S\rangle_{f=0} = \left(\frac{2 - 3/4 - 3/4}{2}\right) - \left(\frac{-3/4 - 3/4}{2}\right) = 1, \tag{38}$$

This means that the energy difference between the $f = 1$ and $f = 0$ states is given by

$$\Delta E_{\text{hfs}} = \frac{2\alpha^4 m_e g_e g_I}{3}\frac{m_e}{m_N}. \tag{39}$$

We return to this result in part e) when we calculate the energy of the photon corresponding to this transition.

d) For $\ell \neq 0$, we cannot use the argument of spherical symmetry to simplify the calculation, so we have to calculate the expectation value of the entire HFS perturbation. Using the relation provided (see also Eq. (20)):

$$\langle H_{\text{hfs}} \rangle = \frac{\alpha g_I}{2m_N m_e} \langle \mathbf{I} \cdot \mathbf{A} \rangle = \frac{\alpha g_I}{2m_N m_e} \frac{\langle \mathbf{I} \cdot \mathbf{J} \rangle}{j(j+1)} \langle \mathbf{A} \cdot \mathbf{J} \rangle. \tag{40}$$

Let us expand the second factor of the right-hand side:

$$\langle \mathbf{A} \cdot \mathbf{J} \rangle = \left\langle \frac{\mathbf{L} \cdot \mathbf{J}}{r^3} + \frac{g_e}{2} \left(\frac{1}{r^3} \left(3(\hat{\mathbf{r}})(\mathbf{S} \cdot \hat{\mathbf{r}}) - \mathbf{S}\right) + \frac{8\pi}{3} \mathbf{S} \delta^3(\mathbf{r}) \right) \cdot \mathbf{J} \right\rangle \tag{41}$$

$$= \left\langle \frac{\mathbf{L} \cdot \mathbf{J}}{r^3} + \frac{g_e}{2} \frac{1}{r^3} \left[3(\hat{\mathbf{r}} \cdot \mathbf{J})(\mathbf{S} \cdot \hat{\mathbf{r}}) - \mathbf{S} \cdot \mathbf{J}\right] \right\rangle, \tag{42}$$

where we have dropped the delta-function term since it only contributes for $\ell = 0$ states. Now, using

$$\mathbf{J} = \mathbf{S} + \mathbf{L}, \tag{43}$$

which helps us in two ways: first,

$$\mathbf{J} \cdot \hat{\mathbf{r}} = \mathbf{S} \cdot \hat{\mathbf{r}} + \mathbf{L} \cdot \hat{\mathbf{r}} = \mathbf{S} \cdot \hat{\mathbf{r}} \tag{44}$$

and second,

$$\mathbf{S} \cdot \mathbf{J} = \mathbf{S}^2 + \mathbf{S} \cdot \mathbf{L} = \frac{3}{4} + \mathbf{S} \cdot \mathbf{L} \quad \text{and} \quad \mathbf{L} \cdot \mathbf{J} = \mathbf{L}^2 + \mathbf{L} \cdot \mathbf{S} = \ell(\ell+1) + \mathbf{S} \cdot \mathbf{L}. \tag{45}$$

where the last terms in each equation cancel when $g_e = 2$.[3] We then have

$$\langle \mathbf{A} \cdot \mathbf{J} \rangle = \left\langle \frac{\ell(\ell+1) - 3/4}{r^3} + 3 \frac{(\mathbf{S} \cdot \hat{\mathbf{r}})^2}{r^3} \right\rangle. \tag{46}$$

All that is left to do is calculate the expectation value of $(\mathbf{S} \cdot \hat{\mathbf{r}})^2$. In tensor notation, we can write it as

$$(\mathbf{S} \cdot \hat{\mathbf{r}})^2 = S_i S_j \hat{r}_i \hat{r}_j = \frac{1}{4} (\sigma_i \sigma_j) \hat{r}_i \hat{r}_j = \frac{1}{4} \left(\delta_{ij} + i\epsilon_{ijk}\sigma_k\right) \hat{r}_i \hat{r}_j = \frac{\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}}{4} = \frac{1}{4}, \tag{47}$$

where we used the fact that $\epsilon_{ijk} \hat{r}_i \hat{r}_j = 0$ since $\hat{r}_i \hat{r}_j$ is symmetric in i and j, while ϵ_{ijk} is antisymmetric. So finally, we can write:

$$\langle \mathbf{A} \cdot \mathbf{J} \rangle = \left(\ell(\ell+1) - \frac{3}{4} + \frac{3}{4} \right) \left\langle \frac{1}{r^3} \right\rangle = \ell(\ell+1) \left\langle \frac{1}{r^3} \right\rangle. \tag{48}$$

Now, the last thing we need is

$$\langle \mathbf{I} \cdot \mathbf{J} \rangle = \left\langle \frac{\mathbf{F}^2 - \mathbf{I}^2 - \mathbf{J}^2}{2} \right\rangle = \frac{1}{2} \left(f(f+1) - i(i+1) - j(j+1) \right), \tag{49}$$

Putting everything together, we have

$$\langle H_{\text{hfs}} \rangle = \frac{\alpha g_I}{2m_N m_e} \frac{\langle \mathbf{I} \cdot \mathbf{J} \rangle}{j(j+1)} \langle \mathbf{A} \cdot \mathbf{J} \rangle \tag{50}$$

$$= \frac{\alpha g_I}{2m_N m_e} \frac{f(f+1) - i(i+1) - j(j+1)}{2j(j+1)} \ell(\ell+1) \left\langle \frac{1}{r^3} \right\rangle. \tag{51}$$

[3] This is intriguing. The nuclear-spin-orbit term here is cancelling part of the electron-spin-nuclear-spin term. Both looked like $\mathbf{L} \cdot \mathbf{S}$, reminiscent of a spin-orbit coupling. One may worry about the *anomalous* magnetic moment of the electron, that is, $\Delta a_e = (g_e - 2)/2 \simeq \alpha/2\pi \simeq 0.00116$, which is a small correction to the electron's magnetic moment that arises from quantum electrodynamics (QED) effects. But clearly, such an effect is of higher order in α (as we will see, HFS is already of order α^4, so this would be a α^5 correction).

Note that implicitly, we have used the fact that $\langle 1/r^3 \rangle$ is the same for all states with the same n and ℓ quantum numbers since the wavefunctions only differ by their angular part. This remains true in our new basis since the transformation from the $|n, \ell, m_\ell, s, m_s\rangle$ basis to the $|n, \ell, j, i, f, m_f\rangle$ basis only mixes states with the same n and ℓ quantum numbers. Equivalently, we can say that the new angular wavefunctions are linear combinations of the good old spherical harmonics, so their angular integration is still normalized.

Using the integral

$$\int_0^\infty dr\, r^2 R_{n\ell}^2(r) \frac{1}{r^3} = \frac{1}{n^3 a_0^3} \frac{1}{\ell(\ell+1/2)(\ell+1)}, \tag{52}$$

we can write the final result as

$$\boxed{\langle H_{\text{hfs}} \rangle = \frac{(Z\alpha)^3 \alpha m_e g_I}{4n^3} \frac{m_e}{m_N} \frac{f(f+1) - i(i+1) - j(j+1)}{j(j+1)} \frac{1}{(\ell+1/2)}.} \tag{53}$$

This is clearly a α^4 correction and it is further suppressed by the ratio of the electron mass to the nuclear mass, which is typically of order 10^{-3} for Hydrogen and even smaller for heavier nuclei. As far as I can tell, nowhere did we assume that $i = 1/2$, so this result should be valid for any nuclear spin. The actual value of g_I will depend on the specific nucleus we are considering. Note that three powers of α come from the Bohr radius (hence, with Z dependence), while the fourth power of α comes from the product of two magnetic moments, which are each proportional to e.

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e) For Hydrogen, $g_I = g_p \simeq 5.6$ and $g_e \simeq 2$. In addition, $m_e \simeq 0.511$ MeV and $m_p \simeq 938$ MeV. The ground state ($\ell = 0$ and $j = 1/2$) has $f = 1$ and $f = 0$ states. Using the result of part c) with $g_e = 2$ and the fact that $\Delta\langle \mathbf{I} \cdot \mathbf{S} \rangle = \langle \mathbf{I} \cdot \mathbf{S} \rangle_{f=1} - \langle \mathbf{I} \cdot \mathbf{S} \rangle_{f=0} = 1$, the energy difference between the two states is

$$\Delta E_{\text{hfs}} = \frac{4\alpha^4 m_e g_I}{3} \frac{m_e}{m_N} \simeq 5.9\,\mu\text{eV}. \tag{54}$$

To convert this energy to Hz, we can use the relation $E = h\nu$, where h is Planck's constant. In natural units, $h = 2\pi$, so we have

$$\nu = \frac{\Delta E_{\text{hfs}}}{h} = 1.4 \text{ GHz}, \tag{55}$$

where we used $\hbar = 1 = 6.58 \times 10^{-16}$ eV s to convert from eV to Hz. The corresponding wavelength is given by $\lambda = c/\nu$, where $c = 1 = 3 \times 10^{10}$ cm/s, so we have

$$\boxed{\lambda = \frac{1}{\nu} \simeq 21 \text{ cm}}. \tag{56}$$

This wavelength is in the radio regime (specifically, in the microwave regime) and is referred to as the H-I line. Detection of this line is a smoking gun for the presence of *neutral* hydrogen. When redshifted, it can be used to detect the presence of neutral hydrogen in the early universe, which is crucial for understanding the epoch of reionization and the formation of the first galaxies. The broadest wavelengths expected in cosmology come from redshifts of $z \sim 20$, which would give us $\lambda \sim 4$ m, still in the radio regime but much more difficult to detect due to the Earth's ionosphere and other sources of radio noise. See, e.g., the EDGES experiment for an attempt at detecting this signal from the epoch of reionization.

The s orbital is the most famous case, but given we did all this work to find what happens for $\ell \neq 0$, let us now investigate what happens for the $2p$ orbital, where $n = 2$ and $\ell = 1$. In particular, in the nuclear spinless case, there are two states depending on how the addition of s and ℓ works out:

$$2p_{j=1/2} \quad \text{and} \quad 2p_{j=3/2}. \tag{57}$$

As we have seen, these states are split by the spin-orbit coupling, and the $j = 1/2$ state is higher in energy than the $j = 3/2$ state. Recall from our fine-structure calculations that

$$E^{\mathrm{FS}}_{n\ell j} = -\frac{m_e \alpha^2}{2n^2}\left(1 + \frac{\alpha^2}{n^2}\left(\frac{n}{j+1/2} - \frac{3}{4}\right)\right), \tag{58}$$

which for $n = 2$ and $\ell = 1$ gives an energy difference (from `mathematica`):

$$\Delta E^{\mathrm{FS}}_{2p} = E^{\mathrm{FS}}_{2p_{1/2}} - E^{\mathrm{FS}}_{2p_{3/2}} \simeq 0.25\frac{\alpha^4 m_e}{8} \simeq 45\,\mu\mathrm{eV} \to \lambda \simeq 2.7 \text{ cm}, \tag{59}$$

which a much larger energy splitting than the $n = 1$ hyperfine splitting (and shorter wavelength for the transition radiation).
Now let us compare with the HFS splitting for the same $n = 2$, $2p$ states. While before we had only two states with $f = 0$ and $f = 1$, including the nuclear spin gives us more combinations: $s = 1/2$ and $\ell = 1$ can give us $j = 1/2$ or $j = 3/2$, and then adding the nuclear spin $i = 1/2$ gives us $f = j \pm 1/2$, so we have four states: $f = 0, 1$ for $j = 1/2$ and $f = 1, 2$ for $j = 3/2$. For the $j = 1/2$ states, we have

$$\begin{aligned}\Delta E^{\mathrm{HFS}}_{2p} &= \langle H_{\mathrm{hfs}}\rangle^{j=1/2}_{f=1} - \langle H_{\mathrm{hfs}}\rangle^{j=1/2}_{f=0} = 0.24\mu\mathrm{eV} \to \lambda \simeq 500 \text{ cm},\\ \Delta E^{\mathrm{HFS}}_{2p} &= \langle H_{\mathrm{hfs}}\rangle^{j=3/2}_{f=2} - \langle H_{\mathrm{hfs}}\rangle^{j=3/2}_{f=1} = 0.10\mu\mathrm{eV} \to \lambda \simeq 1200 \text{ cm},\end{aligned} \tag{60}$$

showing that the HFS splitting of the higher-angular momentum levels are getting smaller and smaller and the wavelengths of the corresponding transitions are getting larger and larger. These wavelengths are deep in the radio regime.

This is an extremely important result for astrophysics. It means that we can detect the presence of neutral hydrogen in the interstellar medium by looking for the 21 cm line in the electromagnetic spectrum.

The drawing in the Voyager Golden Record is the one in the bottom right corner of the image, showing the aligned ($f = 1$) and anti-aligned ($f = 0$) spins of the electron and the proton in the ground state of Hydrogen.

HW 5.2 Solution: this question is solved in Sakurai's book, section 5.3.3. Note that Sakurai uses $e = -0.3$ in natural units, but we will keep the sign of e explicit in our calculations. Here, $e = 0.3$ in natural units.

a) Let us write the Zeeman Hamiltonian explicitly including the spin-orbit coupling term and setting $g_e = 2$. Using $L_z + 2S_z = J_z + S_z$, we can write

$$\hat{H}_{\text{Zeeman}} = \hat{H}_0 + \hat{H}_{\text{int}} = \frac{\hat{\mathbf{p}}^2}{2m_e} - \frac{Z\alpha}{r} + \frac{g_e \alpha}{4m_e^2 r^3}\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} + \frac{e}{2m_e}(J_z + S_z)B. \tag{61}$$

Without the B-field, we already know that n, ℓ, j, m_j are good quantum numbers, so we can write the states as $|n, \ell, j, m_j\rangle$. Now, once we turn on the B field, the Hamiltonian is no longer spherically symmetric. It is easy to see that in that case, j is no longer a good quantum number (J is the full angular momentum that generates rotations), but m_j is still a good quantum number since it is pointing along the B-field and therefore preserves rotations around the z-axis. The picture to have in mind here is that the B-field breaks the 3D spherical symmetry down to a 2D axial symmetry around the axis of the magnetic field. Our new good basis will then be $|n, \ell, m_j\rangle$, which are eigenstates of J_z, S^2, L^2, but not of any of the other angular momentum operators.

We can prove the statements above by computing the commutators of the various operators with the Zeeman Hamiltonian. Let us start with the straightforward ones:

$$[L^2, L_z] = 0, \quad [L^2, S_z] = 0, \quad [S^2, L_z] = 0, \quad [S^2, S_z] = 0, \tag{62}$$

and

$$[L^2, \mathbf{L} \cdot \mathbf{S}] = 0, \quad [S^2, \mathbf{L} \cdot \mathbf{S}] = 0, \tag{63}$$

since L^2 and S^2 are Casimir operators of their respective algebras and $\mathbf{L} \cdot \mathbf{S} = L_x S_x + L_y S_y + L_z S_z$. So, we have proven that l and $s = 1/2$ are good quantum numbers of the entire Zeeman Hamiltonian, even in the presence of the spin-orbit coupling.
Before proceding, let us note the following:

$$\begin{aligned} &[L_z, \mathbf{L} \cdot \mathbf{S}] = i(L_y S_x - L_x S_y) \neq 0, \\ &[S_z, \mathbf{L} \cdot \mathbf{S}] = i(L_x S_y - L_y S_x) = -[L_z, \mathbf{L} \cdot \mathbf{S}] \neq 0, \\ \implies &[J_z, \mathbf{L} \cdot \mathbf{S}] = [L_z + S_z, \mathbf{L} \cdot \mathbf{S}] = 0. \end{aligned} \tag{64}$$

From this we conclude two things: 1) including the spin-orbit coupling, we have that m_ℓ and m_s are not good quantum numbers for the Zeeman Hamiltonian, and 2) the m_j quantum number is a good quantum number for the Zeeman Hamiltonian even in the presence of the spin-orbit coupling, since J_z commutes with the entire Hamiltonian (technically, we also need $[J_z, S_z] = 0$).

We can already note two limits:

1. **Weak B-field regime**: in this case, the spin-orbit coupling is much larger than the Zeeman splitting, so we can treat the B-field term as a perturbation on top of the fine-structure Hamiltonian. In this case, the quantum number j is (almost) a good quantum number and the energy levels are split according to the Zeeman term, which gives us a splitting proportional to m_j.

2. **Strong B-field regime**: in this case, the Zeeman splitting is much larger than the spin-orbit coupling, so we can treat the spin-orbit coupling as a perturbation In this case, the quantum number m_ℓ and m_s are still (almost) good quantum numbers and the energy

levels are split according to the Zeeman term, which we expect to split our states according to $m_\ell + 2m_s$.

b) This is the first case discussed above. We can use the states $|n,\ell,j,m_j\rangle$ as our basis and treat the Zeeman term as a perturbation on top of the fine-structure Hamiltonian. The energy shift is given by

$$\Delta E = \langle n,\ell,j,m_j| \frac{e}{2m_e}(J_z+S_z)B\,|n,\ell,j,m_j\rangle = \frac{eB}{2m_e}\langle n,\ell,j,m_j|\, J_z+S_z\,|n,\ell,j,m_j\rangle\,. \tag{65}$$

The expectation value of J_z is straightforward, but those of S_z are less obvious since the states $|n,\ell,j,m_j\rangle$ are not eigenstates of S_z. Sakurai gives us the answer:

$$\langle S_z\rangle \to \langle n,\ell,j,m_j| \frac{\mathbf{S}\cdot\mathbf{J}}{j(j+1)}J_z\,|n,\ell,j,m_j\rangle = \frac{\langle\mathbf{S}\cdot\mathbf{J}\rangle}{j(j+1)}m_j, \tag{66}$$

where we used the fact that $J_z\,|n,\ell,j,m_j\rangle = m_j\,|n,\ell,j,m_j\rangle$. With $\mathbf{S}\cdot\mathbf{J} = \mathbf{S}^2+\mathbf{S}\cdot\mathbf{L}$, we can write

$$\langle S_z\rangle = \frac{m_j}{j(j+1)}\left(\frac{3}{4}+\langle\mathbf{S}\cdot\mathbf{L}\rangle\right). \tag{67}$$

We already calculated the expectation value of $\mathbf{S}\cdot\mathbf{L}$ in the fine-structure calculation. Using that result, we write

$$\langle S_z\rangle = \frac{m_j}{j(j+1)}\left(\frac{3}{4}+\frac{1}{2}(j(j+1)-\ell(\ell+1)-3/4)\right) = m_j\frac{j(j+1)+3/4-\ell(\ell+1)}{2j(j+1)}. \tag{68}$$

Note that $j=\ell\pm1/2$, so the numerator is always positive, and therefore $\langle S_z\rangle$ has the same sign as m_j. In fact, we can further simplify this expression by plugginhg in both cases of j, which gives

$$\langle S_z\rangle_{j=\ell\pm\frac{1}{2},m_j} = \pm\frac{m_j}{2\ell+1}, \tag{69}$$

which is the result given in Sakurai's book (we used the Projection Theorem to get the expectation value of S_z in the $|n,\ell,j,m_j\rangle$ basis while Sakurai uses the Clebsch-Gordan coefficients).

Putting it all together,

$$\Delta E_{j=\ell\pm\frac{1}{2},m_j} = \frac{eB}{2m_e}\left(m_j+\langle S_z\rangle\right) = \frac{eB}{2m_e}m_j\left(1\pm\frac{1}{2\ell+1}\right) \tag{70}$$

c) In the strong B-field regime, we treat the spin-orbit coupling as a perturbation. As discussed above, the good quantum numbers in this case are approximately $|n,\ell,m_\ell,m_s\rangle$, so the energy shift by our perturbation is easily calculated as

$$\Delta E = \langle n,\ell,m_\ell,m_s| \frac{g_e\alpha}{4m_e^2r^3}\hat{\mathbf{L}}\cdot\hat{\mathbf{S}}\,|n,\ell,m_\ell,m_s\rangle = \frac{g_e\alpha}{4m_e^2}\langle n,\ell,m_\ell|\,\frac{\mathbf{L}}{r^3}\,|n,\ell,m_\ell\rangle\cdot\langle m_s|\,\mathbf{S}\,|m_s\rangle\,. \tag{71}$$

The expectation value of $\mathbf{S}$ is straightforward, since $m_s=\pm1/2$, so we have $\langle m_s|\,\mathbf{S}\,|m_s\rangle = m_s\hat{z}$. The expectation value of $\mathbf{L}/r^3$ is less straightforward, but we can use the fact that $\mathbf{L}$ is a vector operator and therefore its expectation value must be proportional to the only vector in the problem, which is the B-field direction $\hat{z}$, so we can write $\langle n,\ell,m_\ell|\,\mathbf{L}/r^3\,|n,\ell,m_\ell\rangle = \langle n,\ell,m_\ell|\,L_z/r^3\,|n,\ell,m_\ell\rangle\,\hat{z}$ (one can show this a little more carefully with the Wigner-Eckart theorem, but it is not hard to see that the expectation value of L_x/r^3 and L_y/r^3 must be zero by symmetry; if it were not, the orbital angular momentum would have a preferred direction in the x-y plane, which contradicts the symmetry of the problem). Putting both factors together, we have:

$$\Delta E = \frac{g_e\alpha}{4m_e^2}m_sm_\ell\left\langle\frac{1}{r^3}\right\rangle. \tag{72}$$

We have already calculated this in the fine-structure calculation:

$$\Delta E = \frac{g_e \alpha}{4m_e^2} m_s m_\ell \left\langle \frac{1}{r^3} \right\rangle = \frac{g_e \alpha}{4m_e^2} m_s m_\ell \frac{1}{n^3 a_0^3} \frac{1}{\ell(\ell + 1/2)(\ell + 1)}. \tag{73}$$

Finally, we may also include the Zeeman term, for which the energy correction is trivially calculated as

$$\Delta E = \langle n, \ell, m_\ell, m_s | \frac{e}{2m_e}(L_z + 2S_z)B \, | n, \ell, m_\ell, m_s \rangle = \frac{eB}{2m_e}(m_\ell + 2m_s). \tag{74}$$

Putting everything together, we have

$$\Delta E = \frac{eB}{2m_e}(m_\ell + 2m_s) + \frac{\alpha}{2m_e^2} m_s m_\ell \frac{1}{n^3 a_0^3} \frac{1}{\ell(\ell + 1/2)(\ell + 1)}. \tag{75}$$

The spin-orbit coupling in this case is a small correction. Take for example the $n = 2$, $\ell = 1$ state, which has $m_\ell = -1, 0, 1$ and $m_s = \pm 1/2$. The correction is

$$\Delta E_{2p_{3/2}} = \frac{eB}{2m_e}(m_\ell + 2m_s) + \frac{\alpha^4 m_e}{48} m_s m_\ell. \tag{76}$$

So, we can immediately recognize what a strong field here means:

$$\frac{eB}{2m_e} \gg \frac{\alpha^4 m_e}{48} \implies B \gg 0.5 \text{ T}. \tag{77}$$

where we used 1 $\text{eV}^2 \simeq 5.12 \times 10^{-3}$ T to convert from eV to Tesla.

d) For intermediate fields we have no choice other than to find the good basis by diagonalizing the Hamiltonian. The good quantum numbers in this case are n, ℓ, and m_j, so we can write the states as $|n, \ell, m_j\rangle$. Finding the energy levels requires the calculation of Clebsch-Gordan coefficients and matrix diagonalization, which is a bit of a pain but straightforward in principle.